L&S Strategic Plan Data Visualizations

Double Majors by College



File name: double_majors_network_edgebundle_2024-02-29.png

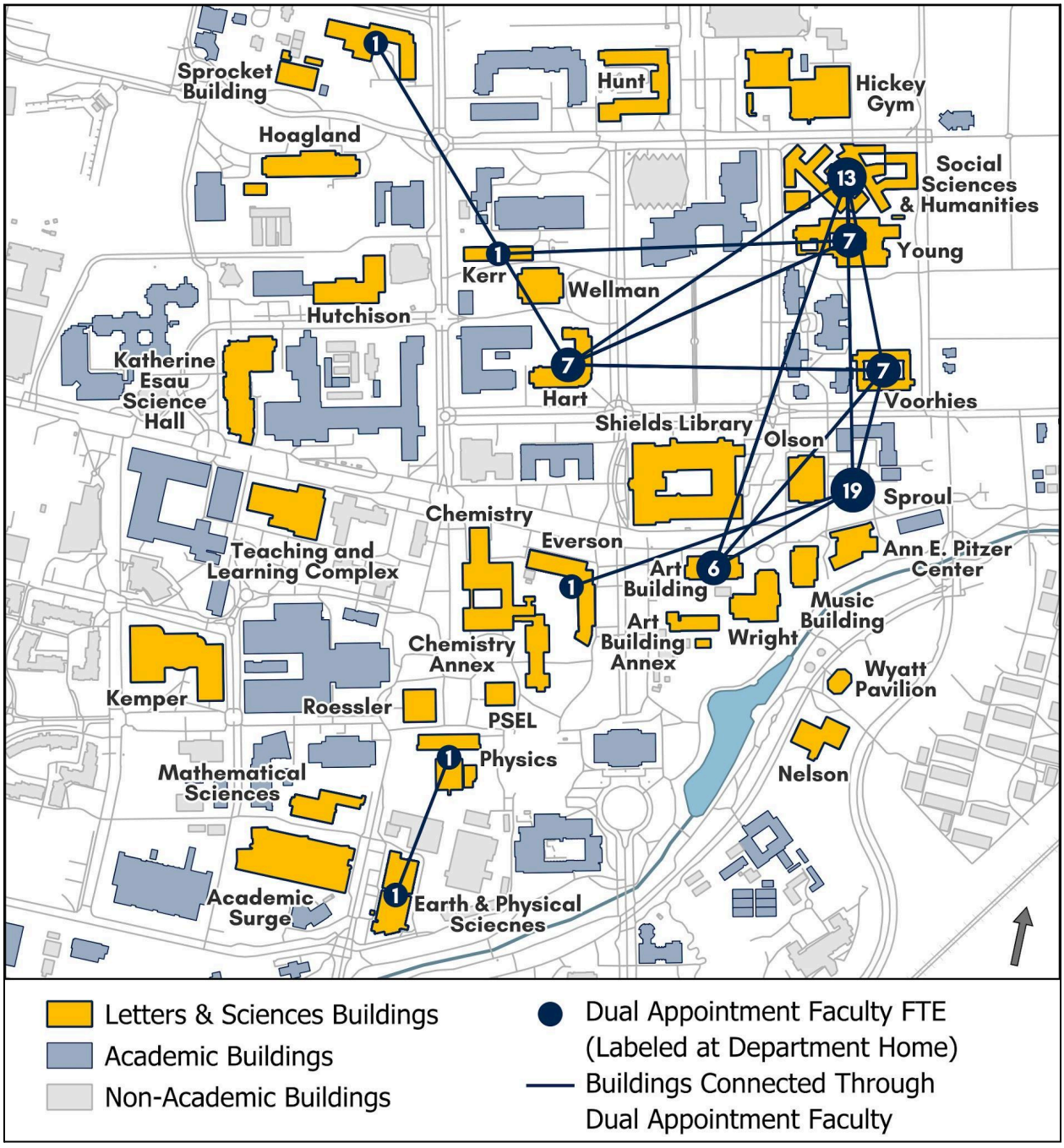
Script: majors.R

Input data: Double Majors all colleges.xlsx

Output data: none

Description: This image connects all of the double majors that contain at least one L&S major with its paired major. Colors represent different colleges: yellow for L&S; dark blue for Ag & Environmental Sciences; gray for Biological Sciences; light blue for Engineering. L&S double majors tend to be within the college. The fewest double majors involve the College of Engineering and (second fewest) Biological Sciences.

Faculty Dual Appointment Map



File name: L&S_Buildings_Dual_Appointments_2024-03-18.tif

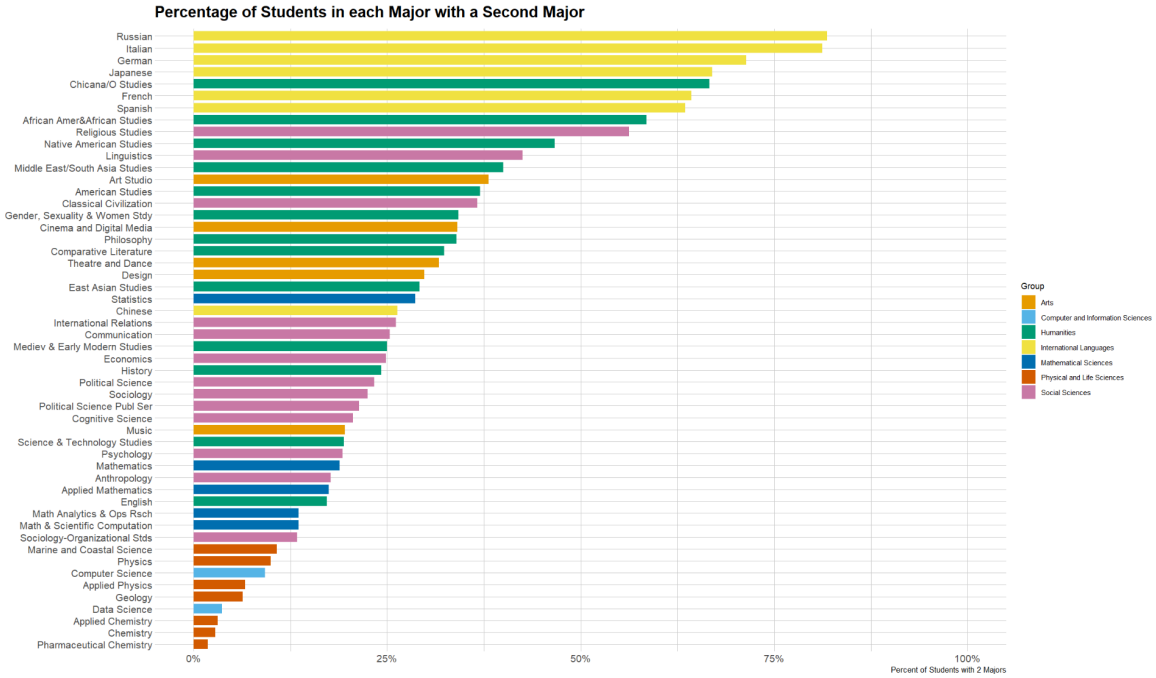
Script: faculty_map.R + cartography work in QGIS

Input data: Copy of Joint faculty appointments.xlsx

Output data: buildings.gpkg

Description: This map illustrates which buildings house L&S faculty, how many dual appointment faculty appointments are in each building, and buildings that are connected by a dual appointment. The units on the numbers shown in the map are appointments (not people). The map shows that the northeast corner of main campus is well-connected with Social Sciences & Humanities and Sproul Hall serving as major connection hubs for the college. L&S is less well connected in the areas of campus that tend to focus on engineering and no connections are present on the Veterinary Medicine campus.

Percentage of Students in Each Major With a Second Major



File name: DoubleMajorPercentages_6-04-2024.png

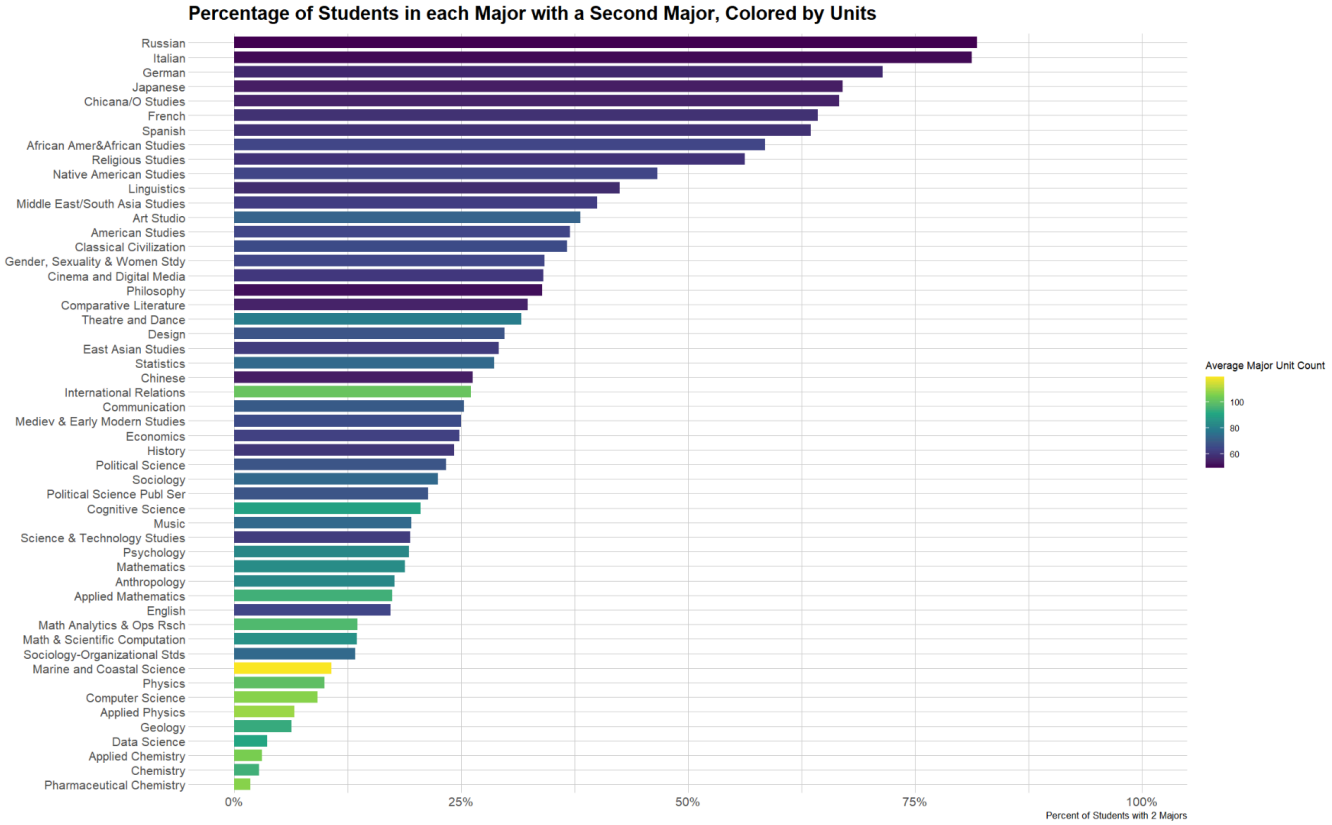
Script: doubleMajorPercentages.R

Input data: Double Majors all colleges.xlsx, Major_Categories5.csv

Output data: N/A

Description: This data shows the percentage of students in each major who have a double major. It is categorized by the group the major belongs in (ex: art, humanities, social sciences, etc.). This data helps put the categories into perspective, and shows that groups such as international languages, and humanities tend to have higher percentages of double majors than math or science majors do overall.

Percentage of Students in Each Major With a Second Major, Colored by Units



File name: DoubleMajorPercentageByUnits_5-21-2024.png

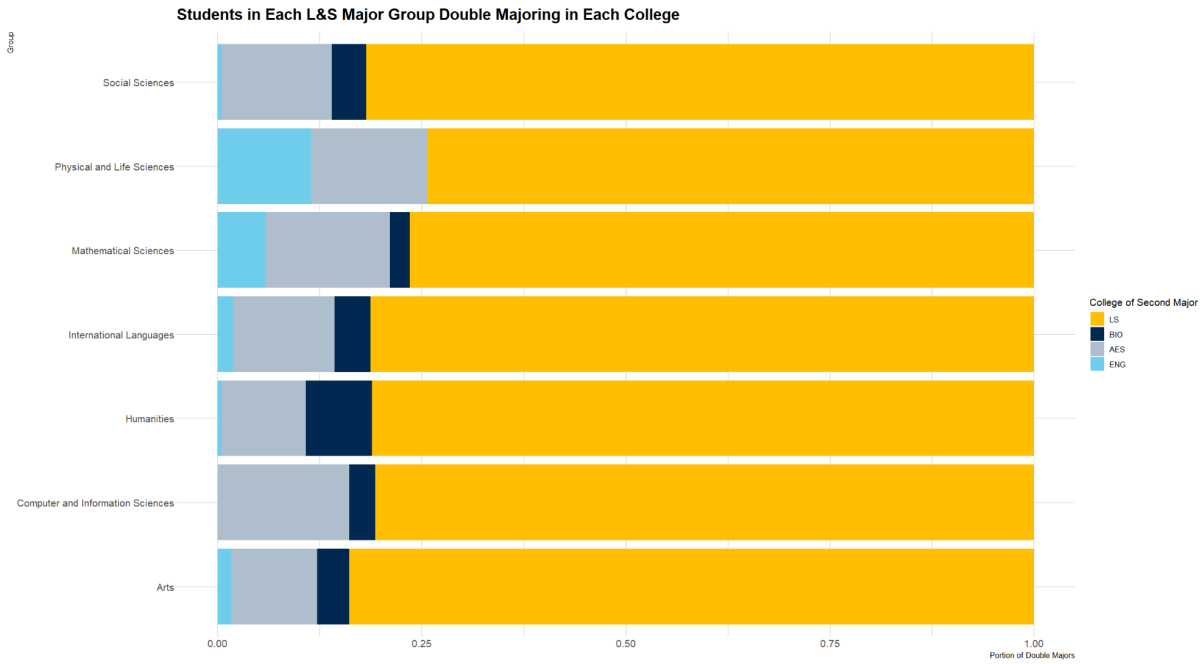
Script: doubleMajorPercentagesUnits.R

Input data: Double Majors all colleges.xlsx, Major_Categories5.csv, programs_units.csv, percent_double_majoring_graphable.csv

Output data: percent_double_majoring_graphable.csv

Description: This data shows a clear correlation between a lower unit count and a greater rate of double majors in any given major. It connects our scraped data of major unit counts with the previous data of double majors. To create this graph, the unit count for each major was averaged within their range (ex: a 100-106 unit major would be graphed at 103), and then BS and BA majors had to be averaged together one more, as our double major data has not been prepared to distinguish or graph BA vs BS students separately. Regardless, the data is clearly telling from this graph.

Students in Each L&S Major Group Double Majoring in Each College



File name: PortionDoubleMajoringInColleges_6-6-2024.png

Script: doubleMajorsColleges.R

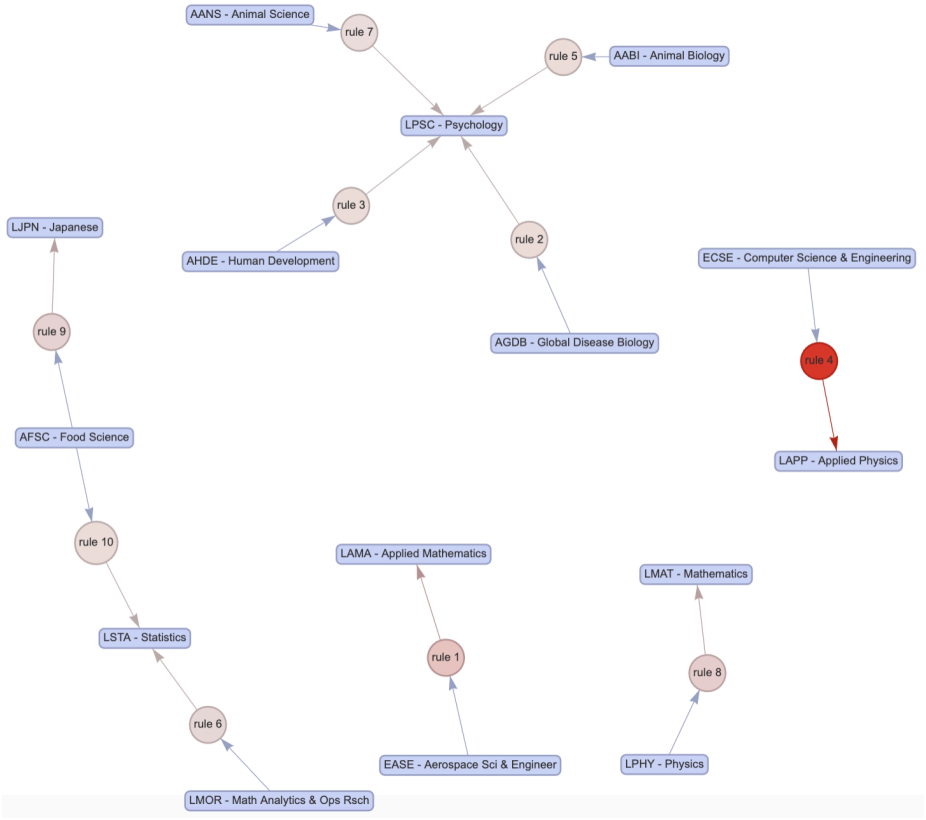
Input data: Double Majors all colleges.xlsx, Major_Categories5.csv

Output data: N/A

Description: This data helps show us where students are double majoring. Across all of L&S, we see that most students are double majoring within the college. However, outside of Letters and Science, we can see that the majority of students across major categories are majoring in Agriculture and Environmental Science (CAES). We can also see trends such as humanities students tending to have a greater chance of a double major in the College of Biological Sciences, or Physical and Life Sciences having a greater percentage of Engineering double majors.

There is also a version of this graph that is zoomed in on only the double majors with a second major outside of L&S, helping zoom in further into that section to see it more clearly *(PortionDoubleMajoringInCollegesNonL&S_5-16-2024.png)*

Prediction for second major based on first major



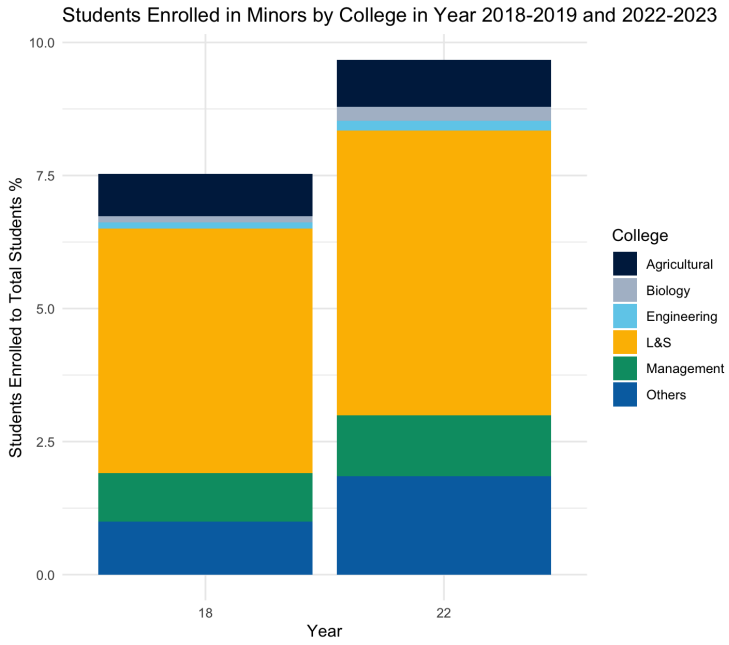
File name: Basket_Analysis_2024-04-26.png

Script: Basket_Analysis.R

Input data: Double Majors all colleges.xlsx - Data.csv

Description: This graph shows the relationship between majors and indicates which major combinations are common. The rules are sorted by confidence in decreasing order and are colored by lift, with darker colors representing higher lift values. A higher confidence value means that the RHS major is very likely to be chosen by students who have already chosen the LHS major(s). Similarly, a higher lift value means that the presence of the LHS major(s) significantly increases the likelihood that the RHS major will also be chosen, compared to the likelihood of choosing the RHS major independently of the LHS major.
Note: data used only included students who have a major in L&S.

Change in Minor Enrollment for 2018-2019 and 2022-2023



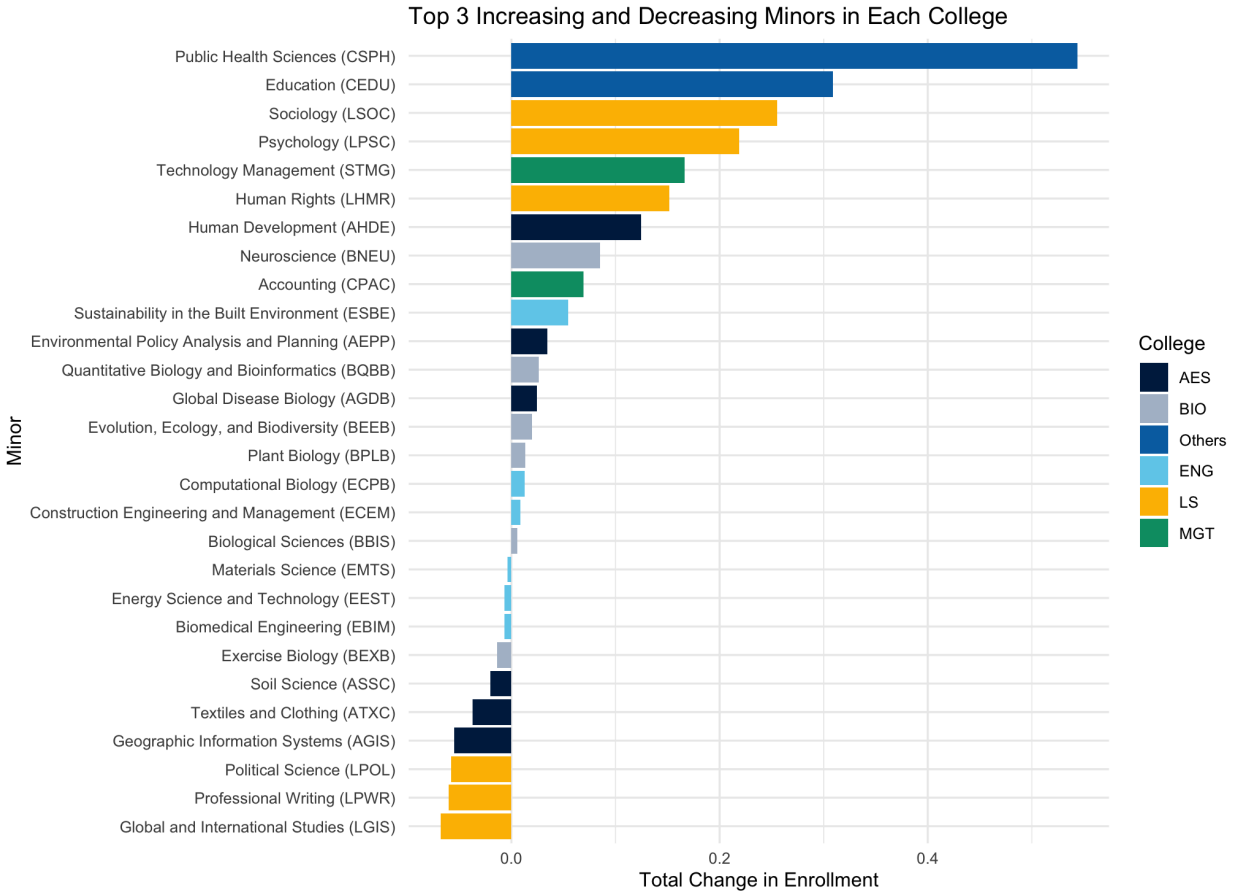
File name: minor_by_year_2026-6.png

Script: minors.R

Input data: Campus Minors.xlsx

Description: This visualization shows us the total growth of the minors between the two years, detailing how each college has contributed to this change and how minors are distributed among them, with L&S having the highest enrollment and Engineering having the least. Although almost all colleges saw an increase, the "others" category, including the minors of Education (CEDU) and Public Health Sciences (CSPH), saw a doubling in enrollment.

Top 3 Increasing and Decreasing Minors in Each College



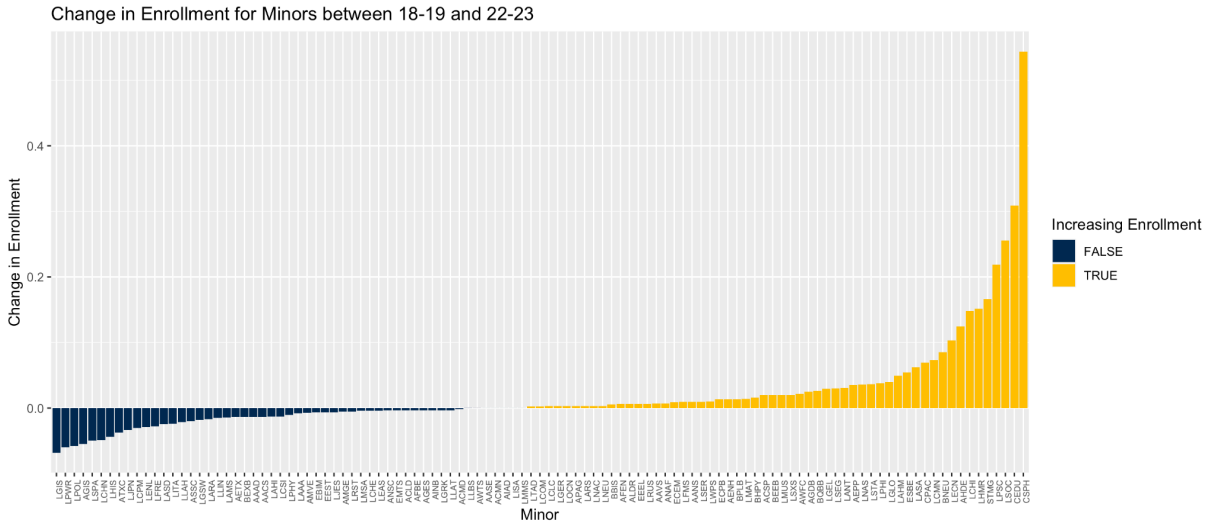
File name: top_3_minors_in_each_college_2024-06-13.png

Script: minors.R

Input data: Campus Minors.xlsx

Description: This visualization shows the top 3 increasing and decreasing minors in each college. Some colleges have less than 6 minors so it is showing all of its minors. It is worth to note that L&S minors had the most decrease in enrollment but also some of the most increasing so it had a lot of changes. However, the college of engineering had the least increasing/decreasing changes. This may show that there's a correlation between the total enrollment in minors in a college and the growth it exhibits.

Growth of Minors Between the Year 2018-2019 and 2022-2023



File name: minor_growth_2024-04-11.png

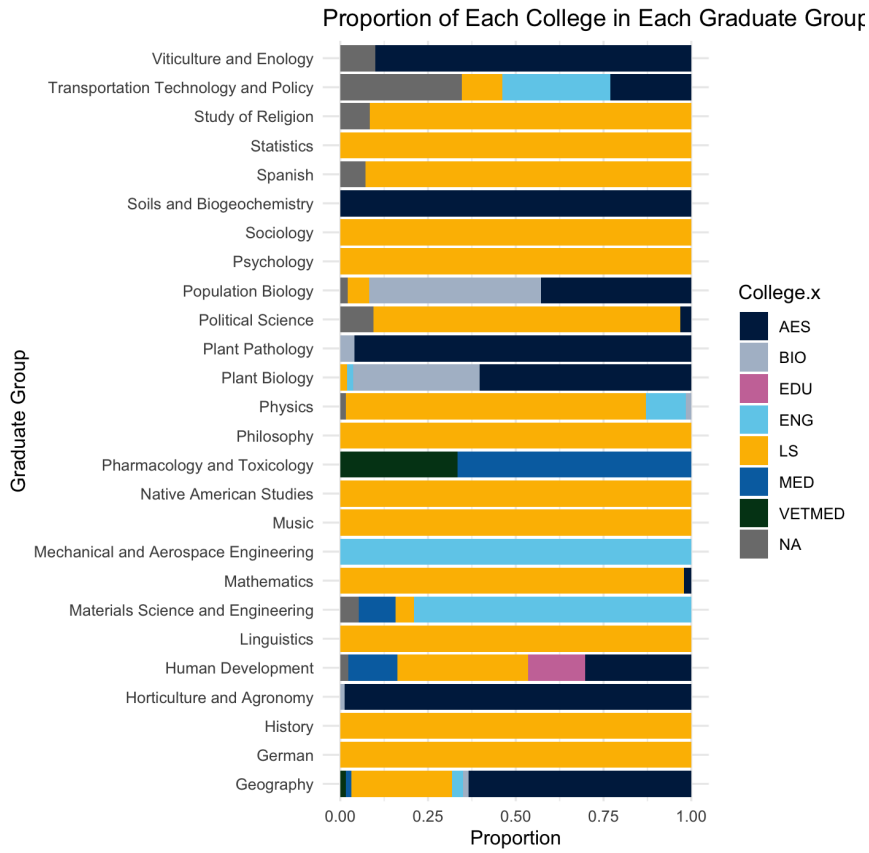
Script: minors.R

Input data: Campus Minors.xlsx

Output data: minor_growth.csv

Description: This visualization shows us the percentage change in enrollments for each minor relative to the total student population at the university, offering insight into how each minor has grown or shrunk compared to the previous year. For example, the minor in Public Health Sciences saw a 0.54% increase in enrollment as a share of the university's total student body between the two years.

Proportion of Each College per last 50 Graduate Group



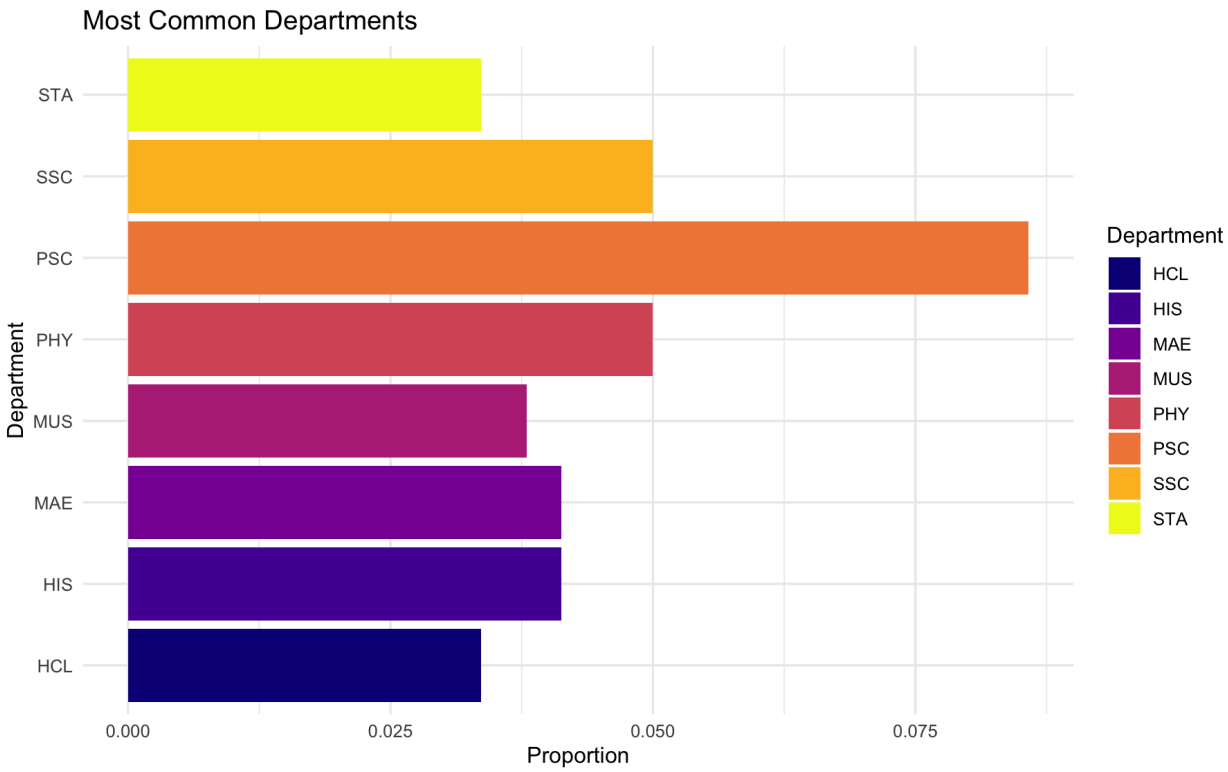
File name: grad-groups-college2-2024-06-12.png

Script: clean_faculty_data.R

Input data: grad_groups.csv

Description: This visualization shows the representation of each college based on the faculty department representation in each graduate group. We can see that a lot of the graduate groups that are based in the College of Letters and Sciences tend to only have faculty from departments in the College of Letters and Sciences as well. STEM, Biology, and Agriculture-focused graduate groups tend to have more representation from other colleges.

Common Departments across Last 50 Graduate Groups



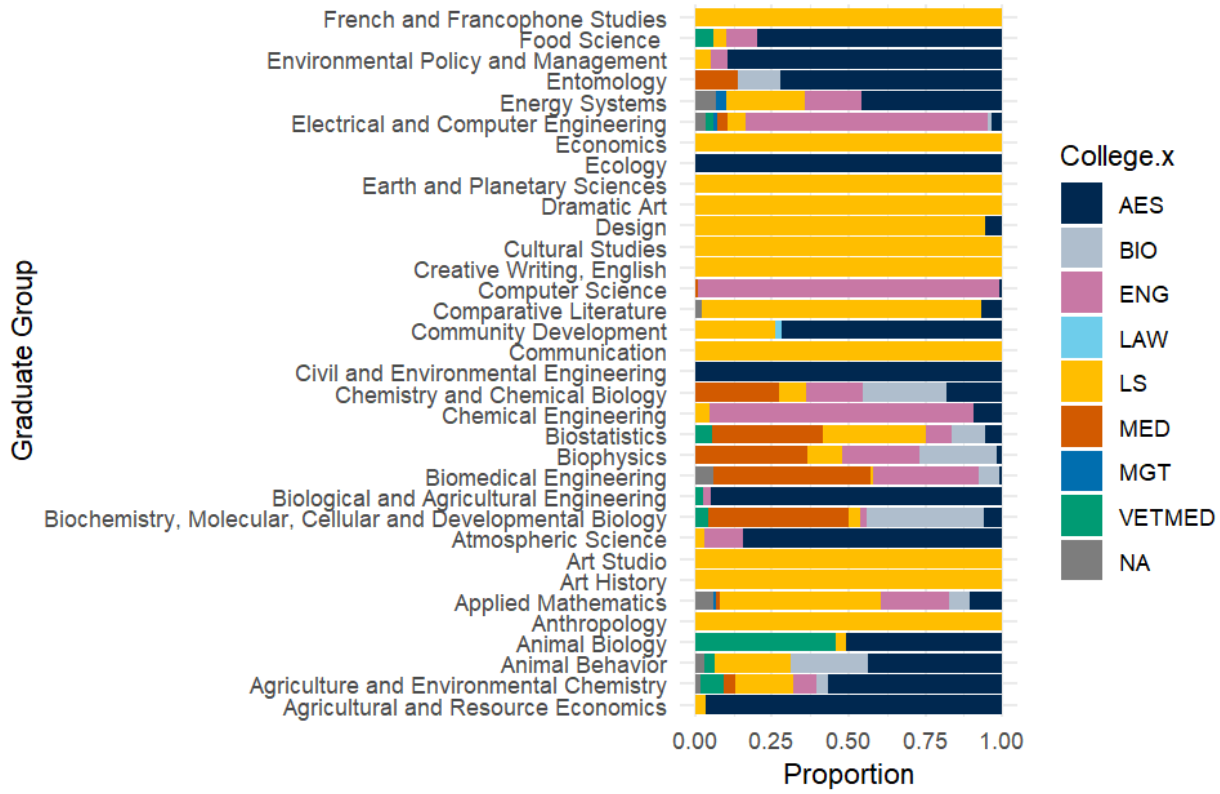
File name: most_common_departments2-2024-06-12.png

Script: clean_faculty_data.R

Input data: grad_groups.csv

Description: This visualization shows top 8 departments represented across the last 50 graduate groups. The top 3 departments, Psychology (PSC), Soils and Biogeochemistry (SSC), and Physics (PHY) make up just around 13 percent of all graduate groups, which could indicate that they have the most potential for interdisciplinary applications.

Proportion of Each College per First 50 Graduate Group



File name: grad-groups-college-2024-06-10.png

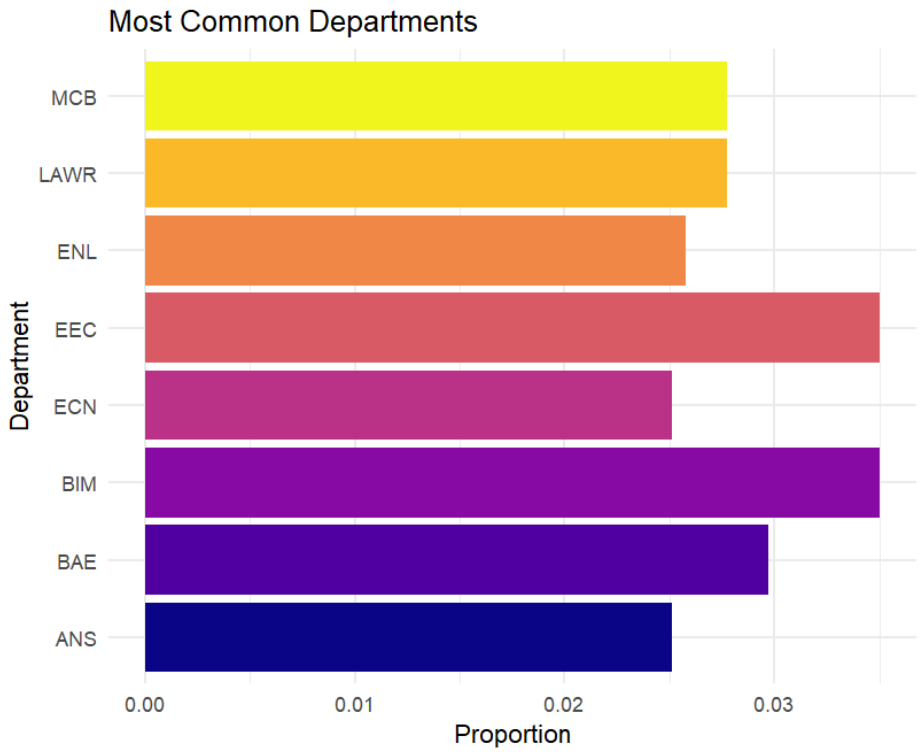
Script: clean_faculty_data.R

Input data: grad_groups.csv

Output data: department_abbreviations.csv, new_departments.csv, first_50.csv

Description: This visualization shows the representation of each college based on the faculty department representation in each graduate group. We can see that a lot of the graduate groups that are based in the College of Letters and Sciences tend to only have faculty from departments in the College of Letters and Sciences as well. STEM, Biology, and Agriculture-focused graduate groups tend to have more representation from other colleges. Note: For this visualization, we only included graduate groups that consisted of at least one faculty from a department in the main college system (L&S, Biological Sciences, Engineering, Agricultural and Environmental Sciences).

Common Departments across First 50 Graduate Groups



File name: most_common_departments.png

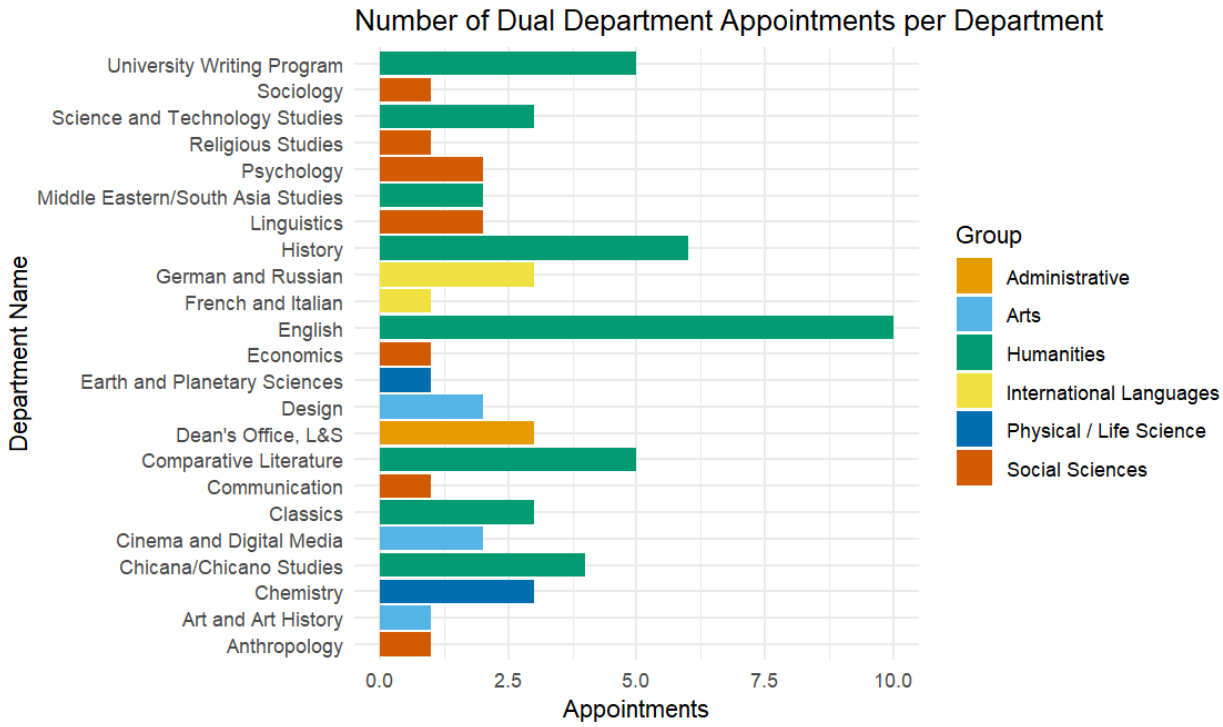
Script: clean_faculty_data.R

Input data: grad_groups.csv

Output data: department_abbreviations.csv, new_departments.csv, first_50.csv

Description: This visualization shows top 8 departments represented across the first 50 graduate groups. The top 2 departments, Biomedical Engineering (BIM) and Electrical and Computer Engineering (EEC), make up just over 3 percent of all graduate groups, which could indicate that they have the most potential for interdisciplinary applications.
Note: For this visualization, we only included graduate groups that consisted of at least one faculty from a department in the main college system (L&S, Biological Sciences, Engineering, Agricultural and Environmental Sciences).

Number of Dual Appointments per Department



File name: dual-appointments-per-dep-2024-06-10.png

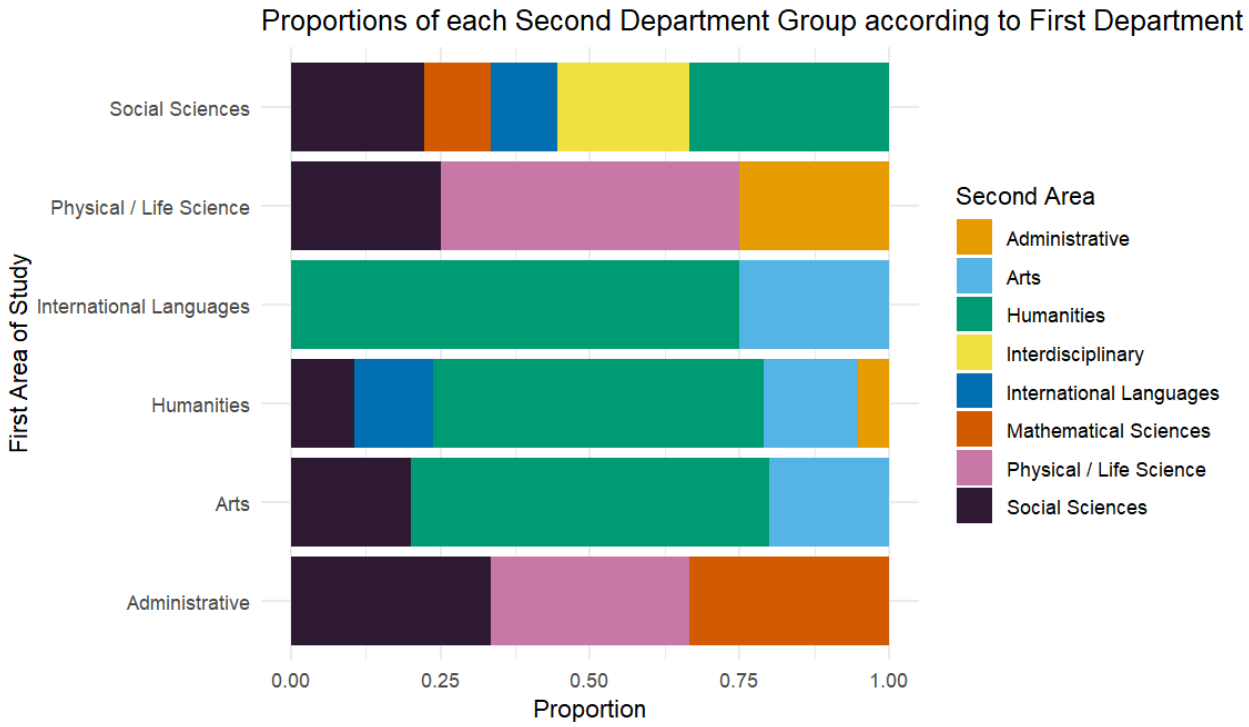
Script: faculty_college.R

Input data: department_college.csv, major_categories_visualized.xlsx, Copy of Joint faculty appointments.xlsx

Output data: faculty_department.csv, LS.csv, faculty_college.csv

Description: This visualization shows the number of faculty, per department, that are additionally appointed to 2 or more departments. They are also sorted into areas of study (i.e. Arts, Humanities, etc). English has the largest number of dual appointed faculty, which could suggest its high interdisciplinary application within the College of Letters and Science.

First & Second Area of Study for Total Appointments



File name: second-dep-group-vs-first-2024-06-10.png

Script: faculty_college.R

Input data: department_college.csv, major_categories_visualized.xlsx, Copy of Joint faculty appointments.xlsx

Output data: faculty_department.csv, LS.csv, faculty_college.csv

Description: This visualization shows the proportion of the second area of study based on the faculty's first area of study. The departments were grouped into areas of study (International Languages, Arts, Humanities, etc.) and the faculty's first and second departments were classified into these areas of study. We can see that the Humanities and Social Sciences departments have the greatest diversity in their dual appointments, showing that they may have more potential for interdisciplinary initiatives in the College Letters and Science.